

Worksheet — 1.5 Legal, Moral, Cultural & Ethical Issues

OCR A Level Computer Science H446 — Component 01, Section 1.5 (1.5.1 legislation; 1.5.2 moral/ethical/cultural/social impact)

Name: _____ Date: _____

Time: ~60 min.

Section A — Skills practice

Activity 1 — Match each law to what it covers

Draw a line, or write the matching letter in the box.

Law		What it covers
1. Data Protection Act 2018 (+ GDPR)	☐	A. How public bodies (police, security services) may monitor/intercept communications, with rules on who can authorise it
2. Computer Misuse Act 1990	☐	B. Intellectual property — protects original works, software, music and designs from being copied without a licence
3. Copyright, Designs & Patents Act 1988	☐	C. How organisations collect, store and process personal data about living individuals
4. Regulation of Investigatory Powers Act 2000 (RIPA)	☐	D. Unauthorised access to computer systems — hacking, malware and unauthorised modification

Answers: 1 = ____ 2 = ____ 3 = ____ 4 = ____

Activity 2 — Which law applies?

For each scenario, name the relevant **Act**. For any Computer Misuse Act scenario, also state **which of the three offences** applies (Offence 1 = unauthorised access; Offence 2 = unauthorised access *with intent* to commit a further offence; Offence 3 = unauthorised *modification*).

1. A company keeps customers' addresses for ten years after they stop being customers, with no reason to do so. Act: _____
2. A student logs in with a teacher's password "just to look around" the staff system, but changes nothing. Act: _____ Offence (if CMA): _____
3. Someone downloads and shares the latest blockbuster film for free from a torrent site. Act: _____
4. A criminal hacks into a bank's servers in order to transfer money into their own account. Act: _____ Offence (if CMA): _____
5. A local council secretly intercepts a resident's emails without obtaining the proper authorisation. Act: _____

6. An attacker releases ransomware that encrypts a hospital's files and demands payment. Act:
_____ Offence (if CMA): _____

Activity 3 — Stakeholders and impacts

Choose **one** ethical issue (e.g. automation in the workplace, mass surveillance, the digital divide, AI bias, offshoring). Write it below, then complete a benefit **and** a drawback for each stakeholder.

Issue chosen: _____

Stakeholder	Benefit (+)	Drawback (-)
Individuals / consumers		
Employees / workers		
Businesses / employers		
Government / wider society		

Activity 4 — Extended-response PLANNING scaffold

Essay question: "The increasing use of **automated decision-making** (e.g. for loan applications, job shortlisting and medical diagnosis) does more harm than good." **Discuss.** (12 marks)

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Plan **before** you write. Fill the skeleton with short bullet points. For each point use: *point* → *consequence* → *stakeholder* → *counter-point*. Aim for **3–4+ stakeholders**.

FOR the statement (arguments it does more harm than good) -

AGAINST the statement (arguments it does more good than harm) -

Stakeholders I will name: _____

Justified conclusion (must take a side — often with a condition): -

Activity 5 — Short-answer definitions

1. **Personal data:** _____

2. **Data subject:** _____

3. **Copyright** (and how does it differ from a **patent**?): _____

4. **Digital divide:** _____

5. **The difference between GDPR and the DPA 2018:** _____

Activity 6 — Balanced argument builder

Scenario: A national supermarket chain announces it will replace **all staffed checkouts** with AI-powered self-checkouts that use **facial recognition** to identify suspected shoplifters automatically.

Examiners repeatedly report that one-sided answers cap out at the middle band. Train the habit here: for **every** stakeholder you must write **one benefit AND one drawback** — no cell may be left blank. Each entry must be a *developed* point (cause → effect), not a single word.

Stakeholder	Benefit (+) of the change	Drawback (-) of the change
Customers / shoppers		
Checkout staff		
The supermarket (business)		
Government / wider society		

Now the evaluative step — answer in one or two sentences each:

(a) Which stakeholder is affected **most seriously**, and why?

(b) Suggest **one safeguard or condition** under which the change would become more acceptable.

Activity 7 — Precision: exactly which law, exactly which offence?

Vague answers such as "data protection laws" or "the hacking law" score **zero** in the exam. For each scenario write the **full name of the Act (including the year)** and, where the Computer Misuse Act applies, the **offence number**. One scenario involves **no offence at all** — spot it and explain why.

#	Scenario	Full Act (+ year)	CMA offence (1/2/3) or "n/a"
1	An employee guesses a colleague's password and reads their private emails "out of curiosity", changing nothing.		
2	A teenager writes and releases a virus that deletes files on infected machines.		
3	A penetration tester, under a signed contract with the company, breaks into its servers to find weaknesses.		
4	An attacker breaks into an online shop's database to copy card details, planning to sell them.		
5	An advertising agency uses a photographer's image from her website in a national campaign without permission or payment.		
6	A quiz app secretly uploads users' entire contact lists to a data broker without telling them.		
7	A government agency intercepts a suspect's phone calls after obtaining the legally required authorisation . Which Act <i>makes this lawful</i> ?		

Which scenario involves **no offence**, and what single word explains why?

Challenge box

A hacker breaks into an online retailer's servers and steals a database of customer card details. The retailer had stored that database **unencrypted** and had kept it for years longer than needed.

(a) Which **two** Acts are relevant here, and **why each**?

(b) Which CMA offence(s) has the hacker committed? Explain.

Section B — Exam practice (30 marks)

Answer **all** questions. Questions marked with an asterisk (*) are marked for the quality of your extended response. Where a question refers to the scenario, your answer **must** be applied to FitPulse — generic answers gain few marks.

Scenario — FitPulse. FitPulse is a UK company that sells fitness wristbands with a companion smartphone app. The app continuously records each user's **heart rate, sleep patterns and GPS location**, and stores this on FitPulse's servers. FitPulse is planning a partnership in which users can share their fitness data with an **insurance company** in exchange for lower premiums; users who refuse will pay the standard (higher) price. Recently: a **hacker** gained access to FitPulse's customer database and viewed records but changed nothing; and a **former employee** has kept a copy of FitPulse's app source code, which she intends to use to launch a rival app.

Q1. State **two** principles of the Data Protection Act 2018 that FitPulse must follow when storing its users' data. [2] (AO1)

Q2. The hacker gained access to FitPulse's customer database and viewed records but changed nothing. State the name of the Act the hacker has broken and identify **which offence** under that Act has been committed. [2] (AO2)

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Q3. Explain why the former employee's plan to reuse FitPulse's source code in a rival app is likely to be unlawful, naming the relevant legislation. **[3]** (AO2)

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Q4. FitPulse's app records users' **GPS location** at all times, and the company is considering selling "anonymised" location histories to advertisers. Explain **two** distinct concerns this raises for FitPulse's users. **[4]** (AO2)

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Q5. The police ask FitPulse to hand over one user's GPS history as part of a criminal investigation. Describe how the **Regulation of Investigatory Powers Act 2000** governs whether and how this data can be obtained. **[3]** (AO2)

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Q6. Some FitPulse customers are elderly or on low incomes and do not own a smartphone capable of running the app, so they cannot access the discounted insurance. Explain how this situation illustrates the **digital divide**, and describe **one** action FitPulse or the insurer could take to reduce its effect. **[4]** (AO2)

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(Total for Section B: 30 marks)